

Comparative Efficacy of 2D and 3D Virtual Reality Games in American Sign Language Learning

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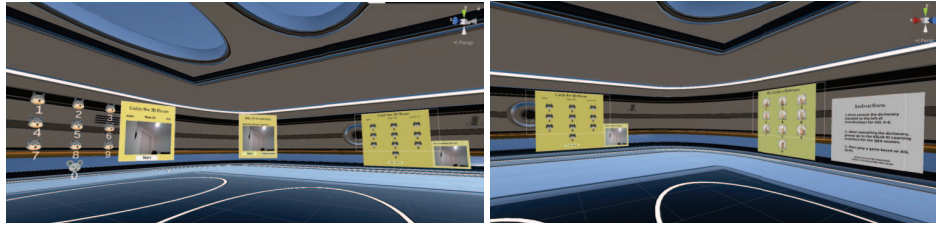


Figure 1: The implemented gamified virtual environments for ASL learning: 3D (left) and 2D (right).

ABSTRACT

With ongoing advancements in virtual reality and gamification, researchers have started exploring their application in sign language learning, aiming at enhancing communication between hearing individuals and the deaf community. This study compares the impact of 2D and 3D games on American Sign Language (ASL) learning, using questionnaires to assess user experience. Results show that 3D games enhance engagement, attractiveness, usability, and efficiency, although user performance remains similar in both environments. The findings suggest the potential of 3D game-based approaches to improve ASL learning experiences while also identifying areas for enhancing their dependability and clarity.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Learning a sign language, not only facilitates communication with the deaf community, but also fosters a more inclusive social environment, demonstrating our dedication to building a society that values diversity and guarantees equal access for all. While current sign language learning initiatives often employ 2D or 3D games, the rationale for selecting one over the other remains unclear [6]. Drawing inspiration from evaluation frameworks employed in related studies involving virtual reality (VR), 3D visualisation, and 2D visual interactions [1], our research identifies a notable gap in evaluating the efficacy of 2D versus 3D environments in sign language learning.

In our study, we developed 2D and 3D versions of a whack-a-mole game that facilitates learning the American Sign Language (ASL) numbers from 0 to 9 and conducted a user study to compare the two versions. The interface has common elements with prior work by the authors presented in [5], where only a 2D environment was studied, and in [7], which was built around user-defined gestures. We adopted the methodologies proposed in Bangor *et al.* [2] and Schrepp *et al.* [3], and performance scores, aiming at answering the

research question: “*What is the impact of game type (2D versus 3D) on system usability, user experience, and learning outcomes in ASL learning?*”. Our main contribution is the collection of empirical evidence suggesting that 3D game environments positively impact user engagement and experience, particularly in terms of attractiveness, usability, and efficiency.

2 IMPLEMENTATION

The 3D VR game for learning the ASL numbers 0-9 is shown in Figure 1 (left). Users interact with the scene utilising the eye-tracking functionality of HTC Vive Pro. When a user’s attention is fixed for three seconds, the system enables them to click and choose an object. A built-in camera was utilised for image acquisition, connected to a PC with openCV. Gesture recognition utilised Mediapipe [9], an open-source platform compatible with standard PC cameras.

In both 2D and 3D versions, the game interface consists of a grid-like layout with the locations numbered 0-9. Players earn one point for correctly signing the location of the gopher, indicating their understanding of ASL gestures. To maintain an engaging experience, we implemented a time limit of 3 seconds for each round. If the player does not sign the correct location within the given time, a new gopher automatically appears, providing the player with another opportunity to score points. The game’s duration is a manageable 30 seconds, which was deemed enough for the players to demonstrate their ASL skills.

3 METHODOLOGY

The 24 participants aged 23-34 years ($M = 28.13, SD = 2.55$) were willing to communicate with deaf people but had no hearing impairment. None of them had any prior experience or formal education in ASL, meaning that the study targeted individuals with no prior sign language knowledge. After being briefed on the purpose of the study, participants signed consent forms. The whole procedure followed the Ethics Guidance of Durham University’s ethics committee.

Aiming at enhancing the learning process by combining instructional scaffolding approaches [8], we divided the learning process into three stages: 1) **Learn**: Participants were encouraged to navigate and explore the virtual learning platform at their own pace, familiarising themselves with the layout and function of each user interface. Following this, they were asked to spend three minutes familiarising themselves with the 0-9 ASL expressions, utilising the 0-9 ASL dictionary interface. 2) **Practice**: To enhance comprehension of 0-9 ASL, the participants were asked to engage in a Q&A session on the Quiz interface, for three minutes, during which they

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were presented with randomly generated questions to respond to. 3) **Assess**: After the participants completed the first two stages, we presented them with the whack-a-mole game. First, the participants ($N_{2D} = 12, N_{3D} = 12$) were asked to play the game once (30 seconds) and to complete the system usability scale questionnaire (about 1 minute) and the user experience questionnaire (about 3 minutes). In the second attempt at the game, half of the participants continued in the same environment, while the other half swapped between 2D and 3D learning environments. Thus, we had four groups in total, of six participants each: **2D-2D**, **2D-3D**, **3D-2D**, and **3D-3D**. The experimental design was inspired by the replicate cross-over designs used in medical research [4].

Three dimensions of the VR learning environments were evaluated: 1) **Usability**, using the well-established System Usability Scale (SUS) questionnaire [2]. 2) **User experience**, using the user survey method of Schrepp *et al.* [3], comprising six scales: *Attractiveness*, *Efficiency*, *Perspicuity*, *Dependability*, *Stimulation*, *Novelty*. 3) **User performance**, analysing the participants' game scores as quantitative indicators of the success of the learning process.

4 RESULTS

Comparing Usability: Both 2D and 3D learning environments demonstrated high levels of usability, achieving scores of 76.25 ($SD = 5.15$) and 80.63 ($SD = 3.41$), respectively, on the 0-100 SUS scale. These scores fall within the "good" usability range, as defined in Bangor *et al.* [2]. The low standard deviations suggest that the majority of participants had consistently positive experiences with both interfaces. These findings highlight the success of creating user-friendly interfaces for both the 2D and 3D learning environments.

Comparing User Experience: Figure 2 highlights the superior user experience of the 3D version compared to the 2D version. Both received positive ratings across various scales, but the 3D version outperformed in *attractiveness*, *efficiency*, *novelty*, *perspicuity*, and *stimulation*. The participants favoured the 3D version for its immersive and visually innovative experience.

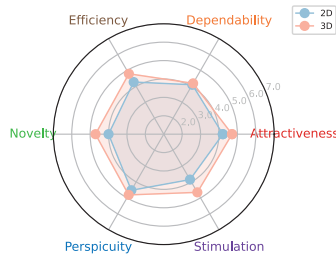


Figure 2: Comparing 2D and 3D on the six scales of user experience.

The comparison suggests that the added dimensionality and immersive elements of the 3D version contributed to a superior user experience, however, as Figure 2 indicates, both were perceived as having low dependability. Qualitative feedback indicated that while the participants thought that both versions met their basic needs, there were areas where they would like to see improvements, such as the complexity of the game and the difficulty of completing the levels. Despite the strengths of the 3D version, it is worth noting that the 2D version still received positive ratings across all scales. Overall, the results show the importance of considering a variety of factors when designing game environments for user experience.

Comparing User Performance: First, we analyse score differences over the two environments on the participants' first attempt. These scores correspond to the first column of Table 1, and we compare the first two rows against the last two. The *t*-test yielded a *p*-value of 0.43, suggesting no significant difference between the versions despite the 2D's marginally higher mean score (10.16 vs 9.50).

Table 1: Score means and standard deviations, for the four groups, for the first (I) and the second (II) attempts.

Group	mean (I)	s.d. (I)	mean (II)	s.d. (II)
2D-2D	10.50	2.22	17.33	2.92
2D-3D	9.83	1.34	14.67	2.49
3D-2D	9.33	1.70	14.83	1.95
3D-3D	9.67	1.49	16.00	2.58

Next, to assess the general learning effect, we compare scores between first and second attempt. The *t*-test indicates a significant learning effect ($p < 0.001$), with the mean score increasing from 9.83 to 15.70 between the two attempts. The experiment's symmetric design ensures the result's independence from the order of playing the 2D and 3D versions. Additionally, increased standard deviations in the second attempt across all groups, as seen in Table 1, indicate a non-uniform learning effect.

To study the effect of familiarisation with the VR environment, as distinct from the general learning effect, we compared the second attempt scores (third column of Table 1) between participants who swapped environments (2D-3D and 3D-2D) and those who did not (2D-2D and 3D-3D). The corresponding means were 14.75 and 16.66, and the *t*-test showed a small yet statistically significant ($p = 0.043$) detrimental effect of swapping VR environments.

5 CONCLUSION

We provided initial evidence of 2D and 3D VR games' effectiveness in enhancing ASL number learning. Notably, the 3D version elicits higher ratings in *attractiveness*, *usability*, and *efficiency*, compared to the 2D version, highlighting its potential to enhance user engagement. Nevertheless, refinements are required to enhance its dependability and clarity. Future studies should investigate design elements affecting user experience in 2D and 3D versions and enhance generalisability by diversifying participants, increasing sample size, and broadening learning content.

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